

Southdown/CEMEX Plant Upgrade Research

- Upgrade Description Source: [EPA's 2007 NOV](#)
 - The upgrades were documented in a Southdown 'Authorization for Expenditure' (AFE) #98097 related to the 'Lyons Capacity Increase Project' at a cost of \$10.3 Million.
 - Attached to AFE # 98097 is a description of the Lyons Plant Expansion project. This description lists several significant changes that were components of the expansion project including:
 - Raw Mill System - increase production rate from 130 to 140 tons per hour by upgrading the existing raw material driver circuit or adding an additional flash dryer circuit.
 - Kiln System - upgrade cooler drag chains, upgrade cooler fans, and upgrade cooler grates.
 - Finish Mill System - install a high efficiency separator to increase production by 10% on the mill system, install a new baghouse for the separator in order to maximize use of the new separator.
 - Oxygen Plant - purchase two 50 ton per day oxygen plants, one in 1998 and one in 1999.
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Raw Mill Systems in Cement Production

Description of CEMEX's Raw Mill System Upgrade in the EPA Notice of Violation (2007):

"increase production rate from 130 to 140 tons per hour by upgrading the existing raw material driver circuit or adding an additional flash dryer circuit"

Raw mill systems are key components in cement production. They process the raw materials that are eventually converted into cement clinker in the kiln. Typically, raw materials like limestone, clay, and sand are mixed and ground into a fine powder known as "raw meal," which is then preheated and fed into a rotary kiln for clinkerization.

Components of Raw Mill Systems

- Feeding System: Belt conveyors and elevators transport raw material to the mill.
- Grinding Unit: Consists of rollers and a table that pulverize the material.
- Separator: Separates the ground material based on particle size.
- Dust Collectors: Remove dust and fine particles from the system.
- Material Transport: Fans, air slides, and elevators move materials between units.
- Flash Dryer Circuit: Used for drying wet materials before grinding (not always present).

Upgrading a raw mill system to increase its production rate from 130 to 140 tons per hour involves several considerations:

- **Motor Capacity:** The drive motor might need to be upgraded to handle the additional load. **If the motor has to be upgraded, you might need a larger unit. This could require additional space, typically in the range of a few square meters.**

- **Material Handling/Raw Material Driver Circuit:** Conveyors, elevators, and feeders may also need to be upgraded or optimized. Upgraded conveyors, elevators, and feeders might require more space, although the increment is usually modest. **However, if a new line is added, it can occupy a significant amount of space.**
- **Grinding Unit:** Increasing the efficiency of the grinding unit by using higher efficiency separators, increasing roller pressures, or optimizing the grinding media. Typically, upgrades to existing grinding units don't significantly change the unit's footprint unless a new mill is added.
- **Flash Dryer:** If moisture content is an issue, adding an additional flash dryer circuit could speed up the drying process, making grinding more efficient. **Adding an additional flash dryer circuit could be significant in terms of space requirements. Flash dryers vary in size but expect to allocate a few hundred square meters for a new unit, including necessary auxiliary systems. A reasonable estimate for a complete new flash dryer circuit, with all accompanying upgrades, might be in the ballpark of several hundred to over a thousand additional square meters, plus vertical expansions.**
- **Dust Collection:** More throughput might require enhanced dust collection and possibly a more robust waste material handling system. A new or upgraded dust collector may require additional floor and vertical space. **Dust collectors can be sizable, sometimes requiring an area upwards of 50 square meters and considerable vertical space for ductwork.**
- **Controls and Instrumentation:** System controls may also need to be updated to handle the increased throughput. Additional or upgraded control systems could necessitate a modest increase in control room space or new installations on the plant floor.
- **Utilities:** Upgraded or additional electrical systems, compressed air systems, and cooling systems might also require additional space, which can vary significantly based on requirements.
- **Potential Permits Required on a County Level:**
 - Building Permits: For any new construction or significant modifications.
 - Air Quality Permits: For any changes that affect emissions.
 - Waste Disposal Permits: If the changes result in different types or quantities of waste.
 - Fire Safety Permits: Due to increased volumes of raw materials, which could be flammable or pose other risks.
 - Noise Level Permits: If the upgrade results in increased noise levels.
 - Water Use Permits: If the process modification requires additional water.
 - Electrical Permits: For new installations or changes in electrical systems.

Kiln Systems in Cement Production

Description of CEMEX's Kiln System Upgrade in the EPA Notice of Violation (2007):
“upgrade cooler drag chains, upgrade cooler fans, and upgrade cooler grates”

Kiln systems are at the heart of cement manufacturing. They involve the transformation of raw materials into clinker, a primary component in cement. The scale of the upgrade would be quite significant given that it affects the central part of the cement manufacturing process. Below are some details about what an upgrade like the one described above would involve:

- **Cooler Drag Chains:** These are used to transport clinker from the kiln to the cooler. Upgrading could mean using materials that are more heat resistant or that have a higher throughput. Replacements would generally have the same physical dimensions but would be made of better materials.
- **Cooler Fans:** These are critical for the clinker cooling process. More efficient or faster fans can significantly increase the cooling rate, which would be essential to handle the increased production rate. **Depending on the needs, you might need larger fans or perhaps additional fans. This could require some significant changes to the housing or ductwork.**
- **Cooler Grates:** These are part of the cooler where the clinker is spread for cooling. Upgrading could mean better material that is more resistant to the high temperatures, or a design that allows for more efficient heat exchange. Dimensions might remain the same, but new designs could be more intricate, requiring custom manufacturing.
- **Potential Permits Required on a County Level:**
 - Construction Permit: For any structural changes to the facility.
 - Environmental Impact Assessment: Given that you're increasing production, there might be a need to evaluate environmental impacts.
 - Air Quality Permit: Upgrading the cooler fans may affect emissions, so an updated air quality permit may be required.
 - Noise Pollution Permit: If the new equipment is noisier, a new noise pollution assessment and permit might be necessary.
 - Waste Management Permit: If the upgrade involves the disposal of old equipment or construction waste, you may need a special permit for that.

Finish Mill Systems in Cement Production

Description of CEMEX's Finish Mill System Upgrade in the EPA Notice of Violation (2007):
“install a high efficiency separator to increase production by 10% on the mill system, install a new baghouse for the separator in order to maximize use of the new separator.”

Finish mill systems are crucial in the final stages of cement production. They're responsible for grinding the clinker into fine cement powder, and any upgrade to this system would significantly impact the plant's output, energy efficiency, and overall operating costs.

- **High-Efficiency Separator:** This piece of equipment sorts the ground material from the finish mill into fine and coarse fractions. A high-efficiency model could considerably improve the cut size and, consequently, the overall efficiency of the system. High-efficiency separators usually occupy about the same footprint as conventional ones but may require additional peripheral systems, such as advanced control panels and more sophisticated feed mechanisms.
- **New Baghouse for the Separator:** This is an air pollution control device that filters particulates out of the air that passes through. It would be essential for maximizing the efficiency of the new separator. **The size of a new baghouse could be significantly larger if designed to deal with the higher throughput effectively. Expect a sizable structure, possibly multiple stories, depending on the specific requirements.**

Potential Permits Required on a County Level:

- **Construction Permits:** Most likely needed for any physical additions or alterations to the existing plant structures.
- **Environmental Permits:** Including an air quality permit especially because you are introducing a new baghouse and potentially increasing production, and therefore emissions.
- **Electrical Permits:** Given that new, potentially high-powered machinery will be installed, you'll need permits for the electrical work.
- **Health and Safety Permits:** Depending on the jurisdiction, you may need to prove that the new installation does not compromise worker safety.
- **Noise Pollution Permits:** High-efficiency separators and new baghouses could produce more noise, requiring an updated noise assessment.

Oxygen Plants in Cement Production

Description of CEMEX's Oxygen Plant Upgrade in the EPA Notice of Violation (2007):
“purchase two 50 ton per day oxygen plants, one in 1998 and one in 1999”

Oxygen plants in cement production are typically employed to enhance the combustion efficiency in kilns, which can improve clinker quality and reduce energy costs. Let's break down the specific elements you're interested in.

Oxygen plants capable of producing 50 tons per day would be substantial facilities. They would occupy several thousand square feet and likely be 20-30 feet in height.

- **Purpose:** Increasing the oxygen concentration in kiln combustion air can significantly improve the efficiency of the process, possibly reducing fuel consumption and increasing throughput.
- **Integration:** These plants must be fully integrated into the existing process, meaning careful planning around connecting these plants to your kilns and ensuring the safe transfer of oxygen.
- **Safety Measures:** High-purity oxygen is a fire hazard, so **safety measures like fire suppression systems are mandatory.**
- **Control Systems:** Advanced control systems would be required to regulate the flow of oxygen into the kilns, ensuring that the right amount is used at all times.

Potential Permits Required on a County Level:

- **Construction Permits:** You'll need permits for the construction of the new plants, including zoning approvals.
 - **Environmental Permits:** Given that you're making significant changes to your production process, new environmental impact assessments may be required. This may include air and water quality permits.
 - **Safety Permits:** The storage and use of large amounts of oxygen will require specialized safety permits and possibly also a risk assessment.
 - **Utility Permits:** For any new utilities being installed (e.g., electrical systems to power the plants).
 - **Operational Permits:** You'll likely need a specific permit for the operation of an oxygen production facility, which could be classified as a hazardous operation.
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